

Train Reservation System

Group: Shane Dias (10179)
Serene Demello (10181)
Jacell Jamble (10189)

Branch: SE COMPS-A

Batch: A

Data Structure: Linked List

Description: The program is a Train Reservation System implemented in C, designed to manage passenger seat bookings dynamically using linked lists. It defines a Reservation structure containing passenger names, seat numbers, and a pointer to the next reservation. Key functionalities include creating new reservations with dynamic memory allocation and adding them to the linked list for efficient management. The program can also delete Reservations based on seat number. The program aims to simplify operations like booking and viewing seat reservations in a scalable way while demonstrating fundamental data structure concepts.

Output:

```
Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 1
Enter passenger name: Shane
Enter seat number: 7
Reservation added for Shane at seat 7.

Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 1
Enter passenger name: Serene
Enter seat number: 11
Reservation added for Serene at seat 11.

Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 1
Enter passenger name: Jacell
Enter seat number: 19
Reservation added for Jacell at seat 19.

Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 3
Reservations:
Seat 7: Shane
Seat 11: Serene
Seat 19: Jacell
```

```
Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 2
Enter seat number to cancel: 7
Reservation at seat 7 has been cancelled.
```

```
Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 2
Enter seat number to cancel: 11
Reservation at seat 11 has been cancelled.
```

```
Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 2
Enter seat number to cancel: 1
No reservation found for seat 1.
```

```
Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 2
Enter seat number to cancel: 19
Reservation at seat 19 has been cancelled.
```

```
Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 3
No reservations made.
```

```
Train Reservation System
1. Add Reservation
2. Cancel Reservation
3. Display All Reservations
4. Exit
Enter your choice: 4
Exiting system...
```